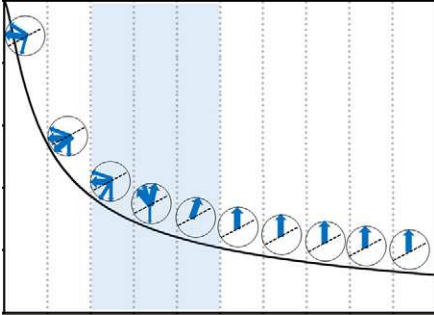


Original Research



Core Ideas

- Pressure variations of a steep hillslope were monitored with an in situ tensiometer network.
- Vertical flows appeared when pressure variations were negligible.
- Lateral flows were primarily associated with transitional periods of pressure.
- The soil moisture characteristic curve was characterized in the context of flow path variation.

Y. Gwak and S. Kim, Dep. of Environmental Engineering, Pusan National Univ., Jangjun-dong san 30 Kumsungku, Pusan 609-735, Korea (ROK); Y. Gwak, Disaster Prevention Research Division, National Disaster Management Research Institute, 365, Jongga-ro, Jung-gu, Ulsan, 44538, Korea (ROK); K. Lee, Dep. of Civil Engineering, Daegu Univ., Gyeongsan, Gyeongbuk, Korea (ROK); S. Noh, Dep. of Civil Engineering, Univ. of Texas, Arlington, TX 76019; H. An, Dep. of Agricultural and Rural Engineering, Chungnam National Univ., Deahak-ro 99, Yuseong-gu, Daejeon 305-764, Korea (ROK). *Corresponding author (kimsangh@pusan.ac.kr).

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Flow Vector Analysis Methods for a Pressure Sensor Network on a Steep Hillslope

Yongseok Gwak, Sanghyun Kim,* Khilha Lee, Seongjin Noh, and Hyunuk An

Investigation of the temporal variations in flow regimes is important for understanding the hydrological processes of hillslopes. An in situ tensiometer network was designed and installed to monitor pressure variations of a steep hillslope. Based on this network, two methods, one using the total potential gradient and another using the centroid potential gradient, were proposed to estimate the flow gradient vectors. A flow gradient vector can be separated into vertical, lateral, and transverse components, which were applied to characterize three-dimensional pressure fields. Time series of the potential hydraulic gradients were obtained at two different depths within the regolith during a 4-mo period. The fine temporal resolution of the pressure gradient vectors allowed detailed analysis of the transitional flow paths that occurred during the wetting and drying stages of rainfall events. The transitional behavior of the flow angles in various temporal rainfall distributions was assessed to characterize distinct flow regimes in both the saturated and unsaturated zones. Vertical flows (i.e., downward and upward flows) appeared when pressure variations were negligible, whereas lateral flows (i.e., diagonal recharge and upslope lateral flow) were primarily associated with transitional periods of pressure. The flow vectors rotated from upward to downward flow in a clockwise direction during wetting periods and from downward to upward flow in drying periods. The impact of macropore and root development, density of gravel, and weathered rock with fracture in depth can be expressed through distinct probability density distributions of flow angles between the upper and lower pressure networks. Flow vector analysis with respect to soil moisture measurements indicated that the soil moisture characteristic curve could be characterized based on the context of the flow path variation at a point on a steep hillslope.

Abbreviations: SMCC, soil moisture characteristic curve.

Hillslope hydrological processes have been explored through studies of flow paths and runoff generation mechanisms (Horton, 1933; Dunne and Black, 1970; Buttle and McDonald, 2002; Martini et al., 2015; Penna et al., 2015). Various factors have been investigated to improve the understanding of these soil water processes, such as the configuration of residence times, connectivity, and the distribution of spatial saturation patterns (Torres et al., 1998; Uchida et al., 2005). The primary factor driving the movement of soil water is the spatial distribution of the total potential, which consists of the gravitational force and the soil suction potential that results from capillary forces affected by evapotranspiration and infiltration (McCord and Stephens, 1987; Jury and Horton, 2004).

The two primary flow components on a hillslope, i.e., the vertical and lateral flows, depend on the soil wetness gradient or soil water potential following rainfall events, the topography, and other soil properties such as texture and macropore density (Harr, 1977; Lu et al., 2011; Lv et al., 2013). Vertical flow dominates in homogenous soil environments under both saturated and unsaturated conditions, whereas lateral flows (e.g., subsurface flow and overland return flow) appear in the presence of inclined bedrock, soil layering, and a heterogeneous soil matrix (Dunne and Black, 1970; Zaslavsky and Sinai, 1981; Miyazaki,

1988; Warrick et al., 1997; Reid, 1997; Retter et al., 2006). The generation of lateral flows is often associated with the development of preferential flow or perched water tables (Haria and Shand, 2006). Under unsaturated conditions, the generation of lateral flow depends on more factors than it does under saturated condition because evapotranspiration, hysteresis, heterogeneity, and anisotropic hydraulic conductivity are all strongly associated with soil water tension (McCord and Stephens, 1987; McCord et al., 1991; Torres et al., 1998; Tromp-van Meerveld and McDonnell, 2006; Sinai and Dirksen, 2006).

Characterization of flow variations at the hillslope scale began with a theoretical study of unsaturated flow generation supported by field measurements (Weyman, 1973) and field tension measurements for two-dimensional flow analysis (Harr, 1977). The additional use of tracers provided visual confirmation of flow paths (McCord and Stephens, 1987; Sinai and Dirksen, 2006). Time domain reflectometry was used to detect and measure soil moisture increments caused by the advancement of a wetting front and identified three-dimensional flow vector patterns (Germann and Zimmermann, 2005; Retter et al., 2006). Numerical modeling and measurements of flow indicated that both spatial and temporal distributions of rainfall determine variations in unsaturated flow (Jackson, 1992; Rimon et al., 2011; Lu et al., 2011). Laboratory measurements clarified the role of subsurface topography in the formation of lateral downslope flow profiles based on the angles of subsurface boundaries (Lv et al., 2013). The temporal scale of field pressure measurements has ranged from hours to days (Weyman, 1973; Harr, 1977; Bakker et al., 2007). Numerical approaches based on Richards equation with implementations of preferential flow and dual-permeability models cannot completely handle all types of flow (Šimůnek et al., 2003; Beven and Germann, 2013; Vereecken et al., 2016).

Existing studies of potential gradients have primarily covered lateral downslope and vertical directional flow (Harr, 1977; Lv et al., 2013). Analysis of soil moisture measurements in the study area showed that vertical flow was the primary component of hydrological processes (Kim et al., 2011). Several studies have explored conditions for the generation of vertical and lateral flows using laboratory experiments and numerical approaches (Essig et al., 2009). However, flow convergence and divergence in field hillslopes should be expressed in both lateral and transversal coordinates. Actually, terrain analyses have been used to describe spatial patterns of soil moisture and many other hydrologic variables such as upslope contributing areas and runoff-producing processes (Burt and Butcher, 1985; Moore and Grayson, 1991; Quinn et al., 1991).

The field hillslope condition can be different from the ideal soil condition for capillary flow describing slow water distribution. Preferential flow and its interaction with the soil matrix can make the actual flow more complicated (Beven and Germann, 2013). The vertical distribution of macropores (Beven and Germann, 1982) especially can result in distinct flow direction patterns at different depths.

The recent development of software packages for the simulation of three-dimensional soil water movement (i.e., HYDRUS-3D) was extensive through the implementation of various modules and applications (Šimůnek et al., 2016). However, an application for a three-dimensional model of a field hillslope is still limited largely because of the uncertainty associated with macropores and root development. The impact of the macropore profile to the formation of preferential flow and its secondary flux can be configured through the field measurement of flow angles in three-dimensional flow fields. The degree of soil heterogeneity to the formation of a flow path can be quantified through the probability density distributions of flow angles.

Therefore, the identification of the flow direction can be important not only to characterize the flow regime profile but also to formulate and parameterize an appropriate physically based hillslope hydrological model from various approaches (i.e., the dual-porosity model and the dual-permeability model) through retrieving the soil heterogeneity in terms of the probability density function for flow directions. Furthermore, non-stationary hydrometeorological conditions and soil heterogeneity under in situ conditions result in a wide range of distribution of flow angle in the three-dimensional field, which is temporally more dynamic and highly nonlinear than flow angles simulated in laboratory or numerical studies (Lu et al., 2011; Lv et al., 2013). To understand the features of flow variation along a hillslope, it is necessary to explore the decomposition of the three-dimensional flow vector into vertical, lateral, and transverse components. This analysis can provide answers to questions such as: What is the probability density function for flow direction at the plot scale? How can the details of flow bending be expressed during wetting and drying periods?

The objective of this study was to develop methods for estimating potential gradient vectors using pressure values from the pressure sensor network. Two methods were proposed for this purpose: one to represent the total potential gradient and another to determine the potential gradient at the centroid of the measurement system. The strengths and weaknesses of these methods were analyzed, including an analysis of potential gradients measured during a 4-mo study period in 2010. Furthermore, flow vector analysis in conjunction with field measurements (soil moisture and soil water tension) can address an interdisciplinary issue for hillslope hydrology and soil science: the presentation of hillslope hydrological processes on the soil moisture characteristic curve (SMCC).

Materials and Methods

Study Area

The study area was a steep forested hillslope with granite bedrock located within the Sulmachun catchment basin in northwestern South Korea (Fig. 1). The mean annual rainfall from 2008 to 2012 was approximately 1700 mm, and temperature varied

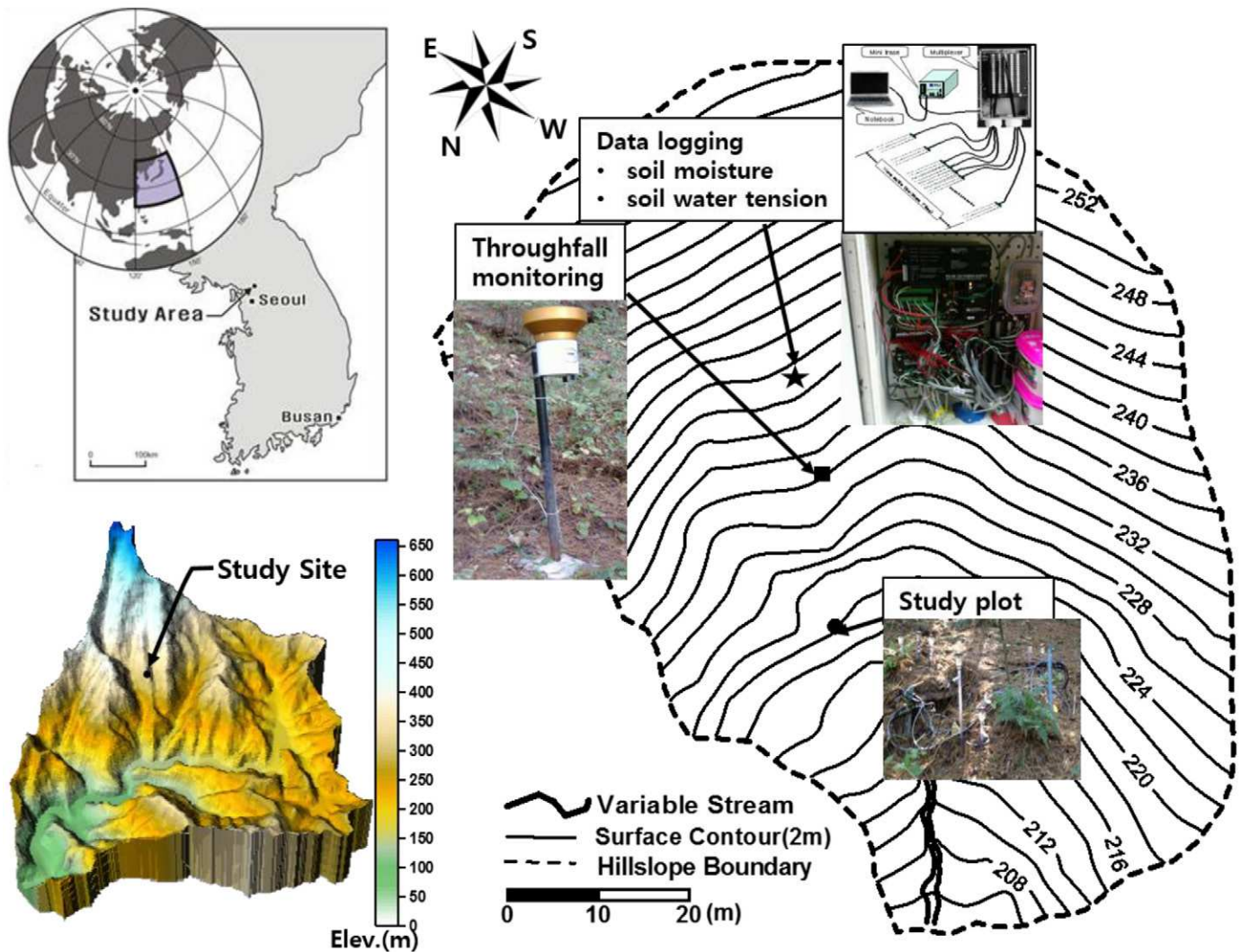


Fig. 1. Location of the hydrometric measurement system with a digital elevation model of the hillslope.

between -15 and 35°C , with a mean annual temperature of approximately 11°C . The areal extent of the study area was approximately 4000 m^2 , with slope angles that ranged between 20 and 35° . The weathered bedrock of the study area consisted of a granitic gneiss with a fractured zone characterized by foliation and joints. Further details of the study area were given by Kim et al. (2011). Soil depths varied approximately between 25 cm at the hilltop to 90 cm in the downslope area. The generation of soil was related to weathering of the granite gneiss. The predominant soil textures consisted mainly of loamy sand and sandy loam ($67.0 \pm 9.1\%$ sand, $30.3 \pm 8.2\%$ silt, and $2.7 \pm 8.2\%$ clay) based on the USDA soil texture standards. The soil analysis was based on measurements at depths of 10 cm at 19 points within the study area and was performed using fine mesh sieves and a laser diffraction particle size analyzer (Beckman Coulter Corp.). A thin organic layer (layer A) at topsoil and subsoil (layer B) were developed to deep depth depending on the depth of soil. An abundance of macropores and root development were found at shallow depths of the soil, and more gravel and weathered rocks

with fractures were found in deeper layers of the soil. The dominant hillslope vegetation was a mixed forest of *Pinus rigida* Mill. and *Solanales*; no systematic vegetation structure was identified across the small scale of the hillslope.

Hydrometric Measurement System

To measure the pressure variations in three dimensions, a pressure measurement system was designed (Fig. 2). Following the definitions of the downslope or upslope directions (Lu et al., 2011), we designated that the directions of lateral flow and vertical flow were parallel to the surface and to gravity, respectively. Tensiometers were installed at depths of 30 , 60 , and 90 cm at three positions arranged in a triangle. Vertical, lateral, and transverse distances between the tensiometers ranged from 30 to 132 cm . The elevation of the two upslope points, points TL (left side) and TR (right side) in Fig. 2, was approximately 30 cm higher than that of the downslope point TB (bottom side). The upper pressure measurement network, which consisted of TL30, TL60, TR30, TR60, and TB30, was used to calculate

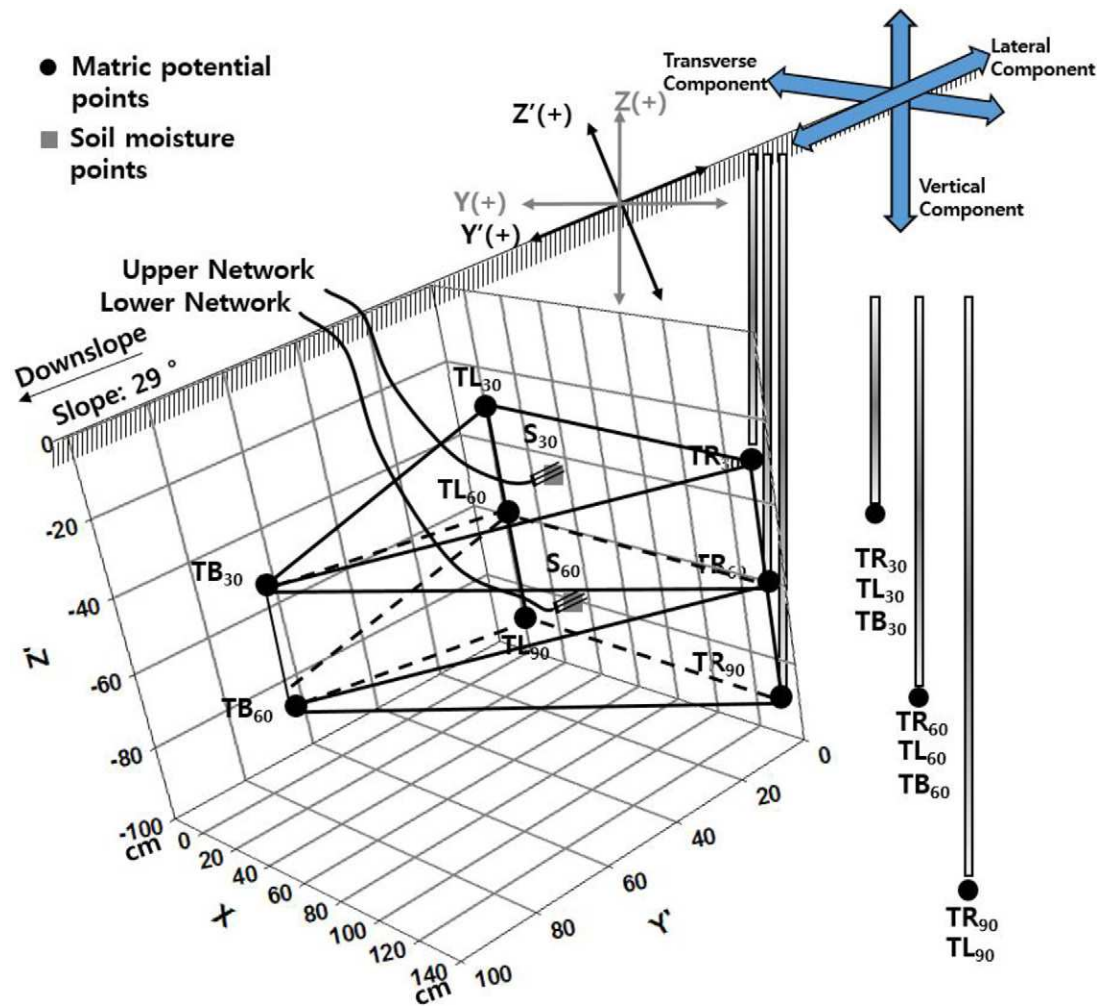


Fig. 2. Schematic of the pressure gradient and soil moisture measurement system of the study area with three flow directions; TL = transducer left; TR = transducer right; TB = transducer bottom; S = soil moisture; 30, 60, 90 = depth (cm) below surface.

the potential gradient in the soil between 30 and 60 cm. The lower pressure network, which consisted of TL60, TL90, TR60, TR90, and TB60, was used to identify the flow gradient vector at depths between 60 and 90 cm (see Fig. 2). Tensiometer rods (SoilMoisture Equipment) were installed in early April 2010, and data were obtained between 30 April and mid-November 2010. Pressure transducers were connected 3 wk after the original installation to stabilize the potential field around the porous cups. To ensure stabilized pressure between the porous cups and the surrounding soil media, only data collected in June to November 2010 were analyzed. Tensiometer transducers GT3-15 and GT3-30 from ICT International were used. These transducers were able to detect pressures within the ranges of -100 to 0 and -100 to 100 kPa, respectively, with a resolution of 0.1 kPa. Soil water tension time series were obtained at 1-min intervals, and moving averages of measurements at 10-min intervals were recorded by a datalogger (CR1000, Campbell Scientific). The soil volumetric soil moisture was measured using MiniTRASE (SoilMoisture Equipment Corp., 2005) every 2 h at depths of 30 and 60 cm (see Fig. 2). A tipping bucket rain gauge (TE525MM,

Campbell Scientific) was used to monitor rainfall events, as shown in Fig. 1.

Delineation of the Potential Gradient

The total soil water potential gradient (i) was used to characterize the potential flow of the soil water based on Richards equation (Šimůnek et al., 2003), even though the impact of preferential flow needs to be properly considered for better characterization (Weiler, 2017). The primary hydrological processes in a hillslope are generated during rainfall and its subsequent period. The total soil water potential included several potentials that generally consisted of the two components for non-swelling soil (Jury and Horton, 2004; Lv et al., 2013),

$$i = \frac{\Delta H}{D} = \frac{\Delta(\varphi_m + \varphi_z)}{D} \quad [1]$$

where i is the total soil water potential gradient, ΔH is the difference in total potentials between two locations [L], φ_m is the soil metric potential measured with a tensiometer [L], φ_z is the

gravitational potential at elevation [L], and D is the distance between two measurements [L].

From the quadrangular pyramid of the pressure gradient measurement system (Fig. 2), two different potential gradient vectors, i.e., the total potential gradient and the centroid potential gradient, were considered (Fig. 3). The total potential gradient represented the summation of all possible potential gradients for Fig. 3a. The total potential gradient can be expressed as three-dimensional vectors, $\mathbf{i}_X$, $\mathbf{i}_{Y'}$ (parallel to the surface), and $\mathbf{i}_Z$ (see Fig. 4a):

$$\begin{aligned}\mathbf{i}_X &= \frac{\sum \Delta \mathbf{H}_X}{4D_X} \\ &= \frac{1}{4D_X} \left[\Delta \mathbf{H}_{12} + \Delta \mathbf{H}_{34} + (\Delta \mathbf{H}_{(15)X} - \Delta \mathbf{H}_{(25)X}) \right. \\ &\quad \left. + (\Delta \mathbf{H}_{(35)X} - \Delta \mathbf{H}_{(45)X}) \right] \quad [2]\end{aligned}$$

$$\begin{aligned}\mathbf{i}_{Y'} &= \frac{\sum \Delta \mathbf{H}_{Y'}}{4D_{Y'}} \\ &= \frac{1}{4D_{Y'}} (\Delta \mathbf{H}_{(15)Y'} + \Delta \mathbf{H}_{(25)Y'} + \Delta \mathbf{H}_{(35)Y'} + \Delta \mathbf{H}_{(45)Y'}) \quad [3]\end{aligned}$$

$$\begin{aligned}\mathbf{i}_Z &= \frac{\sum \Delta \mathbf{H}_Z}{4D_Z} \\ &= \frac{1}{4D_Z} \left[(\Delta \mathbf{H}_{(35)Z} - \Delta \mathbf{H}_{(15)Z}) \right. \\ &\quad \left. + (\Delta \mathbf{H}_{(45)Z} - \Delta \mathbf{H}_{(25)Z}) + \Delta \mathbf{H}_{31} + \Delta \mathbf{H}_{42} \right] \quad [4]\end{aligned}$$

where D_X , $D_{Y'}$, and D_Z represent the distances between pressure sensors in the X , Y' , and Z directions as shown in Fig. 2 and 3. The notation $\Delta \mathbf{H}_{ab}$ represents the different potential differences between locations a and b in Fig. 3a, and the subscripts X , Y' , and Z indicate the converted gradient in the directions of the respective axes from the observed gradient.

The centroid potential gradient was calculated by defining a vector at the point at which the perpendicular lines for the pressure difference vectors calculated with Eq. [2], [3], and [4] could be minimized. Figure 3b illustrates the minimum distances between the pressure sensors $d_{kj(ab)}$, where the first subscript k denotes an axis or multiple axes, and the second subscript $j(ab)$ represents one of the total potential gradients between a and b for each corresponding direction (k). With the impact of the pressure measurement on a centroid potential gradient vector assumed to be inversely proportional to the minimum distance, the general formulation of a vector for Fig. 3b can be expressed as

$$\mathbf{i}_k = \frac{1}{\omega_k D_k} \sum_{j=1}^n \frac{\Delta \mathbf{H}_{j(ab)k}}{d_{kj(ab)}} \quad [5]$$

$$\omega_k = \frac{\sum_{i=1}^n \prod_{j=1}^n a_{ij}}{\prod_{j=1}^n d_{kj(ab)}} \quad [6]$$

$$a_{ij} = d_{kj} \text{ if } i \neq j, \quad a_{ij} = 1 \text{ if } i = j \quad [7]$$

where subscript k represents one of the directions x , y' , or z , n is the number of pressure gradients for each D_k , and ω_k is the weighting factor for each direction [L^{-1}]. Equation [5] can be expressed for the three-dimensional vectors $\mathbf{i}_x$, $\mathbf{i}_{y'}$, and $\mathbf{i}_z$:

$$\begin{aligned}\mathbf{i}_x &= \frac{1}{\omega_x D_x} \left[\frac{\Delta \mathbf{H}_{12}}{d_1} + \frac{\Delta \mathbf{H}_{34}}{d_2} + \left(\frac{\Delta \mathbf{H}_{(15)x}}{d_3} - \frac{\Delta \mathbf{H}_{(25)x}}{d_4} \right) \right. \\ &\quad \left. + \left(\frac{\Delta \mathbf{H}_{(35)x}}{d_5} - \frac{\Delta \mathbf{H}_{(45)x}}{d_6} \right) \right] \quad [8]\end{aligned}$$

$$\mathbf{i}_{y'} = \frac{1}{\omega_{y'} D_{y'}} \left(\frac{\Delta \mathbf{H}_{(15)y'}}{d_3} + \frac{\Delta \mathbf{H}_{(25)y'}}{d_4} + \frac{\Delta \mathbf{H}_{(35)y'}}{d_5} + \frac{\Delta \mathbf{H}_{(45)y'}}{d_6} \right) \quad [9]$$

$$\begin{aligned}\mathbf{i}_z &= \frac{1}{\omega_z D_z} \left[\left(\frac{\Delta \mathbf{H}_{(35)z}}{d_5} - \frac{\Delta \mathbf{H}_{(15)z}}{d_3} \right) \right. \\ &\quad \left. + \left(\frac{\Delta \mathbf{H}_{(45)z}}{d_6} - \frac{\Delta \mathbf{H}_{(25)z}}{d_4} \right) + \frac{\Delta \mathbf{H}_{31}}{d_7} + \frac{\Delta \mathbf{H}_{42}}{d_8} \right] \quad [10]\end{aligned}$$

Evaluating the hillslope hydrological processes as two-dimensional vectors in the downslope direction is useful to identify lateral and vertical flows in runoff generation (Harr, 1977). Two vectors representing the lateral flow and vertical flow, namely $\mathbf{i}_L$ and $\mathbf{i}_V$ (see Fig. 4a), respectively, were identical to $\mathbf{i}_{Y'}$ or $\mathbf{i}_{y'}$ and $\mathbf{i}_Z$ or $\mathbf{i}_z$ depending on the total gradient vector and the centroid gradient vector, respectively. Coordinates for a centroid point in the volume of a quadrangular pyramid could be calculated using the massprop function (identification of a centroid for a three-dimensional objective) in AutoCAD (Autodesk, 2012).

As shown in Fig. 4a, the magnitude of the combined vector for the lateral and vertical flow, $\mathbf{I}_R$, can be expressed as

$$\|\mathbf{I}_R\| = \sqrt{\|\mathbf{I}_V \cos \alpha\|^2 + \|\mathbf{I}_V \sin \alpha + \mathbf{I}_L\|^2} \quad [11]$$

where α is the slope of the regolith.

Following this, the flow angle for $\mathbf{I}_R$ is evaluated as

$$\gamma = \sin^{-1} \left(\frac{\mathbf{I}_L \cos \alpha}{\mathbf{I}_R} \right) \quad [12]$$

Depending on the sign convention for $\mathbf{I}_L$ and $\mathbf{I}_V$, the angle from a positive vertical axis is defined as

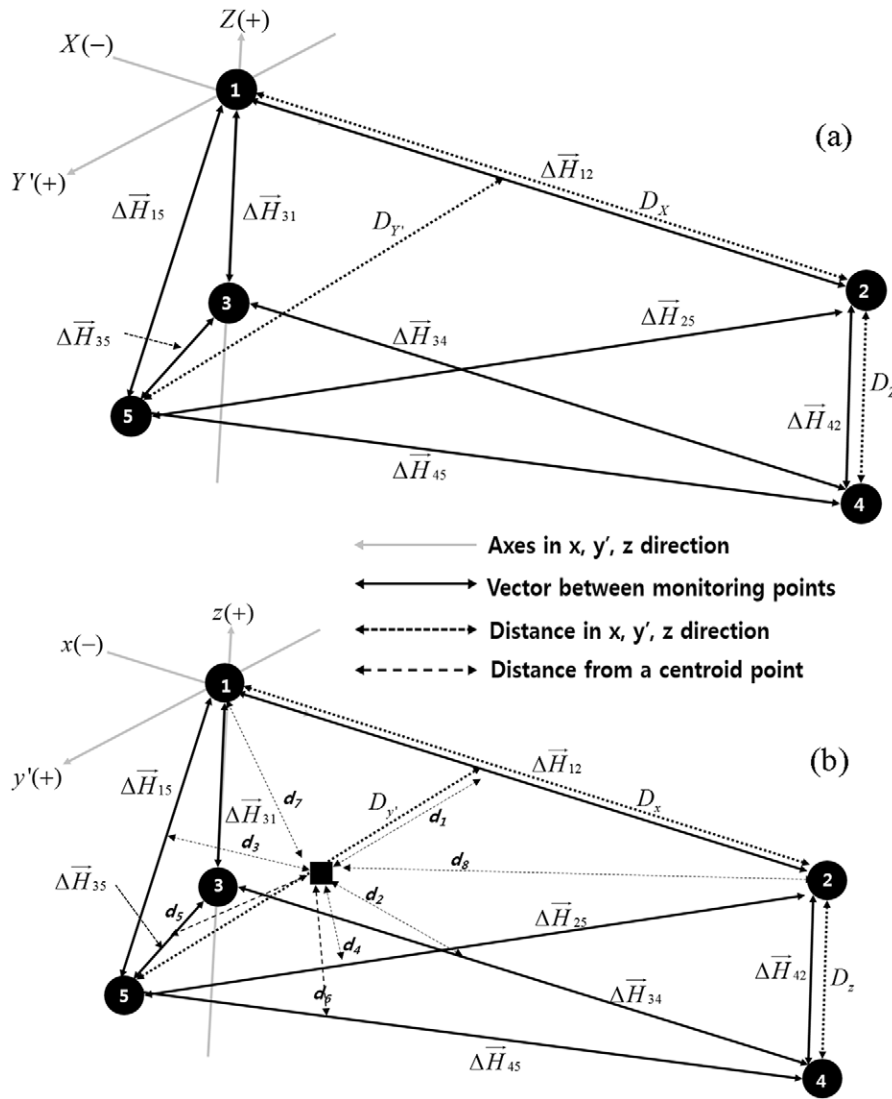


Fig. 3. Schematic of pressure differences (ΔH) and distances (D) used in the evaluation of (a) vectors for total potential gradients and (b) the centroid potential gradient vectors; 1 and 3, 2 and 4, and 5 indicate tensiometers in the left, right, and bottom locations, respectively.

$$\begin{aligned} \theta &= \gamma, \text{ if } \mathbf{I}_L > 0 \text{ and } \mathbf{I}_V > 0 \\ \theta &= \pi - \gamma, \text{ if } \mathbf{I}_L > 0 \text{ and } \mathbf{I}_V < 0 \\ \theta &= \pi + \gamma, \text{ if } \mathbf{I}_L < 0 \text{ and } \mathbf{I}_V > 0 \\ \theta &= 2\pi - \gamma, \text{ if } \mathbf{I}_L < 0 \text{ and } \mathbf{I}_V < 0 \end{aligned} \quad [13]$$

In Eq. [13], the angle θ represents the flow direction between 0 and 360°. Figure 4b presents θ with the surface and flow notations for hydrological processes.

In addition, the resulting magnitude of a three-dimensional gradient vector can be expressed as

$$\|\mathbf{I}_T\| = \sqrt{\|\mathbf{I}_X\|^2 + \|\mathbf{I}_R\|^2} \quad [14]$$

where $\mathbf{I}_X$ is the transverse gradient component.

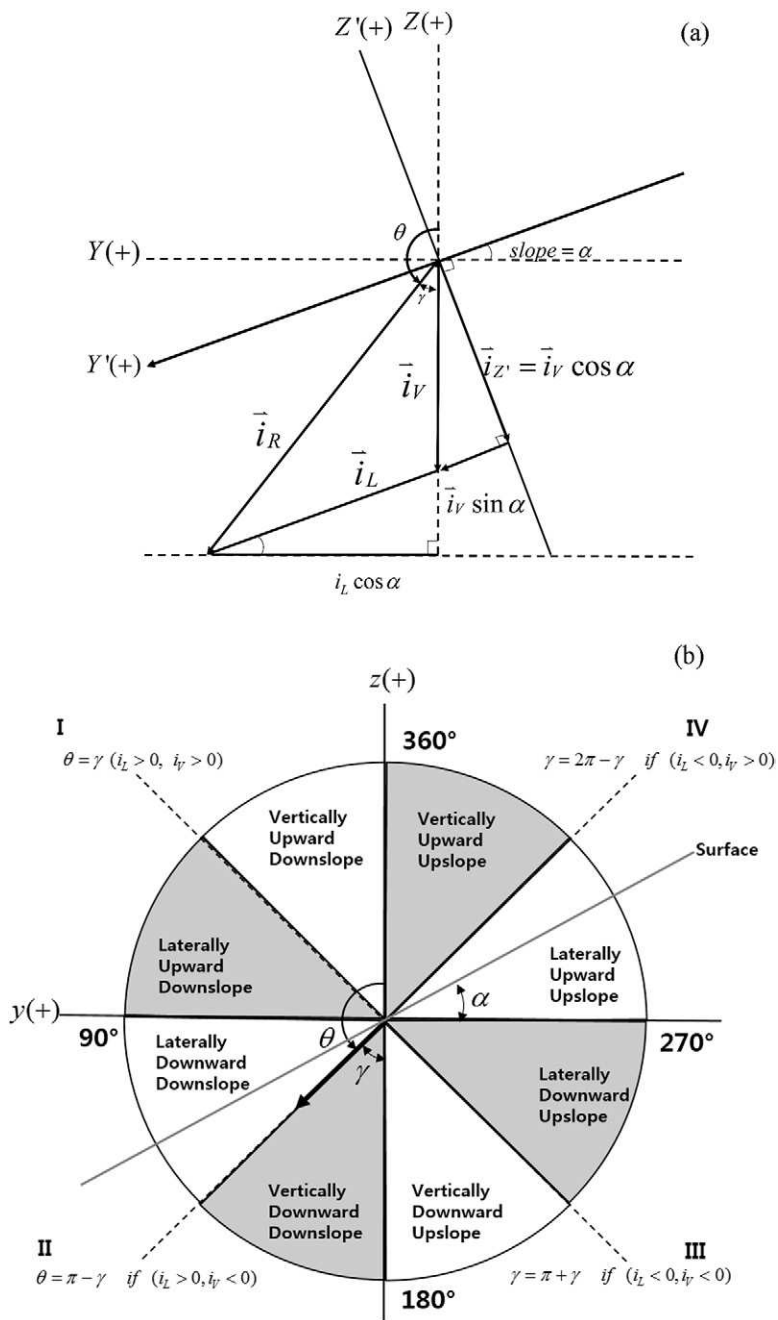
Relationship between Soil Water Tension and Soil Moisture

The SMCC is the relationship between soil water tension and soil moisture. In this study, the van Genuchten model in

RETSC (Windows Version 6.02) was used to delineate a SMCC (van Genuchten et al., 1991). Soil properties (soil composition) determined by field investigations were used to estimate model parameters for the neural network model (Rosetta Lite Version 1.1) in the RETSC program (e.g., residual soil moisture $[\theta_r]$, saturation soil moisture $[\theta_s]$, and parameters for the van Genuchten model such as the curve-fitting parameters n and α). We also programmed the van Genuchten model to evaluate uncertainty associated with measurement errors. The possible error bounds in SMCC can be obtained through a number of iterations of the model, with randomized errors for each measurement equipment.

Results and Discussion

The pressure gradient measurement system, operational in May 2010, collected pressure data between 18 June and 17 Oct. 2010. Figure 5 shows the time series of the rainfall (mm per 10 min) and soil water tension with a positive sign convention ψ (cm), which can be estimated as $\psi = -\psi_{\text{gauge}}$, where ψ_{gauge} is the measured pressure (see Fig. 2). Suction pressure was increased to 500



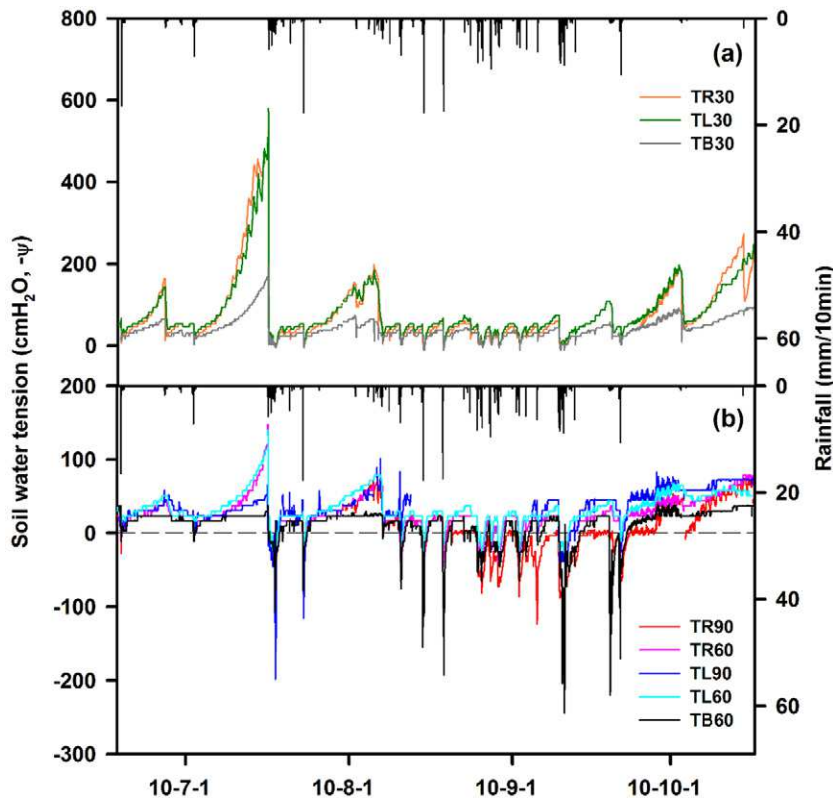


Fig. 5. Time series of rainfall vs. soil water pressure for tensiometers TL (left side), TR (right side), and TB (bottom side) installed at depths of 30, 60, and 90 cm.

Figure 7 presents a frequency analysis of the centroid vector angle for the upper and lower pressure networks in terms of the probability density distributions. Lateral flow in the downslope direction ($60\text{--}90^\circ$) and vertical downward ($130\text{--}185^\circ$) and upward flow ($320\text{--}360^\circ$) were detected in the upper pressure network, whereas the primary flow path for the lower pressure network was vertical downward flow ($130\text{--}180^\circ$). As shown in Fig. 7, the appearance of upslope lateral flow was minimal among the four distinct flow paths.

Soil layering and soil matrix heterogeneity may have been responsible for the lateral flow in the upper network (Miyazaki, 1988; Retter et al., 2006). However, the weathered bedrock in this area was fractured, with foliation and joints that introduced vertical flow into the lower network. Visual inspection revealed an abundance of macropores and the development of root structures in this hillslope (Kim and Jung, 2014). The differences between the upper and lower pressure networks in macropore development and vegetation impact also appeared to be responsible for the pressure vector difference (Tromp-van Meerveld and McDonnell, 2006; Kim, 2012; Beven and Germann, 2013).

The total vector is more applicable for an isotropic measurement network and homogeneous soil properties, whereas the centroid vector is more applicable for a skewed pressure network and a heterogeneous soil environment. Even though the total vector appeared simpler and more feasible regarding computations than

the centroid vector, defining of the mean vector for an estimated soil volume ($= 171,047 \text{ cm}^3$) of the cluster of a tensiometer network seemed to be less meaningful for the configuration of the flow direction. Additionally, the shape of the pressure cluster was not necessarily geometrically symmetrical or matched with the flow regime with any hydrological meaning. However, the centroid vector was a suitable representation of the configuration of hydrological processes at a designated location and depth. The centroid vector also provided better accuracy than did the total vector because the hydraulic gradient was defined for the point that represented the least biased estimate of the flow field of the pressure measurement network.

Transitional Flow in the Pressure Network

The intensity and temporal distribution of rainfall were important factors for determining the variation in the gradient vectors. Figures 8a and 8b show a time series of the transverse, lateral, and vertical components, the two-dimensional angles of the vectors, and the observed soil moisture from the upper pressure network for 26 to 27 June and 16 July 2010, respectively. The accumulated rainfall and rainfall intensities that affected the pressure variation on these dates were 13.6 mm at $0.076 \text{ mm min}^{-1}$ and 52.8 mm at 0.18 mm min^{-1} , respectively, shown in Fig. 8a and 8b. The difference in infiltration depth associated with the difference in rainfall between these two events was responsible for the distinct vector responses.

The vertical components of the centroid vector decreased immediately after the rainfall events; however, the rate of change varied and became negative 3 h and 20 min later for the 27 June event (see Fig. 8a). The lateral component also promptly changed toward the downslope direction; however, it returned to negative values within 50 min, probably because of the small volume of rainfall (Sinai and Dirksen, 2006). In addition, the transverse component showed transitional behavior for 4 h (Fig. 8a). However, the vertical component of the centroid vector for the 16 July event became negative within only 45 min. The lateral and transverse components both decreased and converged into negligible gradients (see Fig. 8b). The difference in the centroid vector responses between the two events was primarily expressed in the vertical vectors (see Fig. 8a and 8b). Variations in the two-dimensional angles appeared to depend on the rainfall intensity and its temporal distribution, with angles varying between 360 and 180° , which indicates a switch from upward flow to downward flow (see Fig. 8a and 8b). The differences in rainfall amount and the temporal distribution were also responsible for the difference in the transition time from upward flow to downward flow between the two events (4 and 1 h; see Fig. 8a and 8b) (Torres et al., 1998).

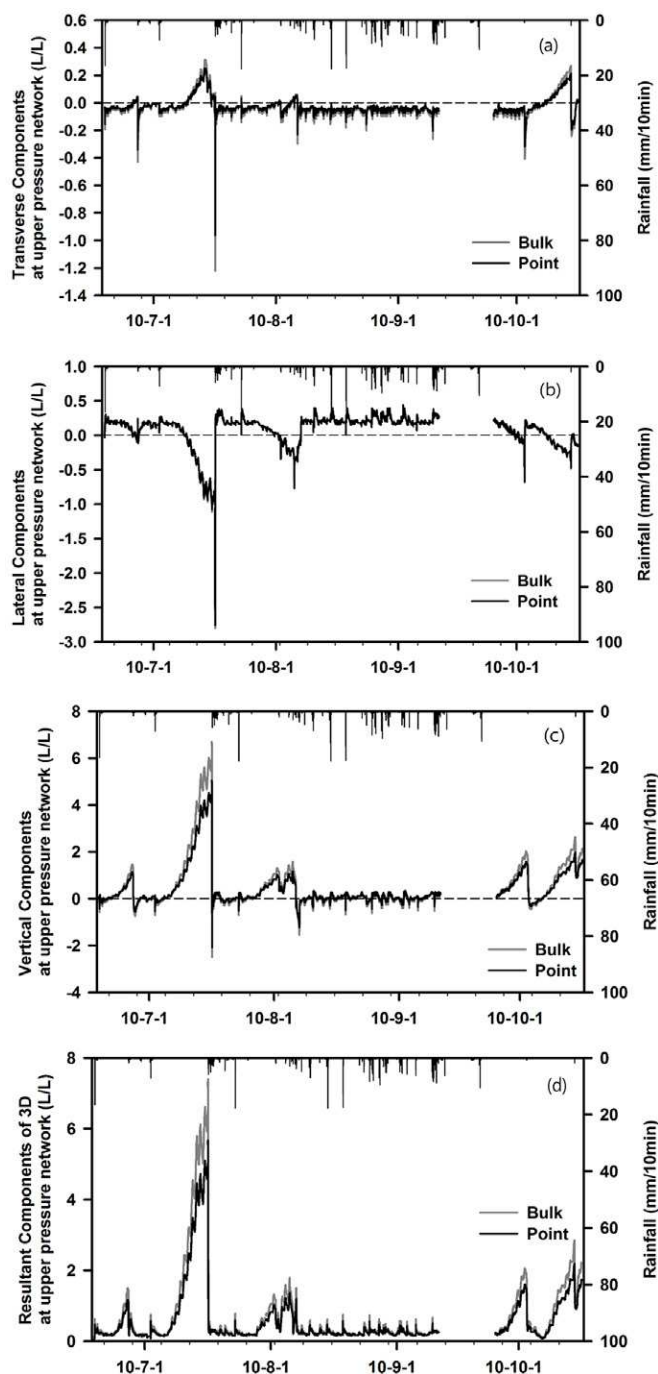


Fig. 6. Time series of the total potential gradient (bulk) and centroid potential gradient (point) vectors from the upper pressure networks comprising tensiometers TL30, TL60, TR30, TR60, and TB30 for (a) transverse components, (b) lateral components, (c) vertical components, and (d) three-dimensional vector magnitudes along with rainfall.

Figures 9a and 9b show the time series of the transverse, lateral, and vertical components, the two-dimensional angles of vectors, and the observed soil moisture from the upper pressure network during drying periods shortly after rainfall events, 18 June and 22 July 2010, respectively. The vertical components of the centroid vectors (Fig. 9a and 9b) responded promptly to the negative

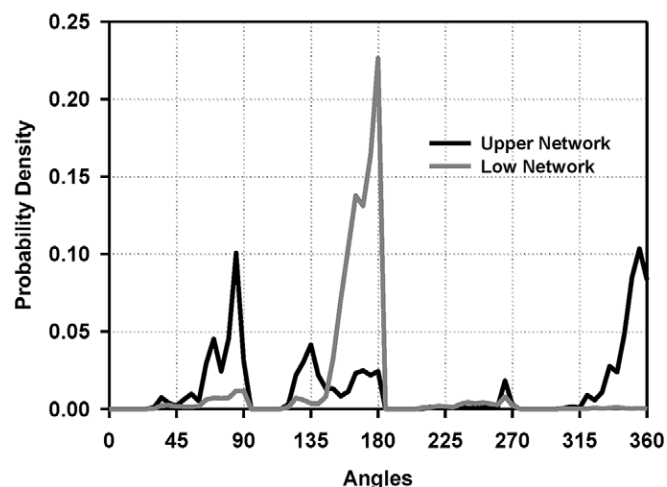


Fig. 7. Flow angle probability density distributions for the upper and lower pressure networks; angles between 315 and 360° are upward flow, angles between 225 and 270° are upslope lateral flow, angles between 135 and 180° are downward flow, and angles between 40 and 90° are downslope lateral flow.

gradients caused by the rainfall events and subsequently returned to positive gradients in approximately 1 d. The positive lateral components in the downslope direction changed into negative gradients over 7 d, while the transverse components showed negligible variation. The temporal variation of the two-dimensional angles and the magnitudes of the resultant vectors were also similar between the two periods (see Fig. 9a and 9b). The angles tended to decrease from 180 to 135° and 90 to 45°, which represented the vertical flow, diagonal recharge (Kim, 2012), lateral flow, and diagonal flux as exfiltration. The timing of the return to vertical upward flow (360°) was approximately 5 d for the two periods, as shown in Fig. 9a and 9b. The change in flow direction after rainfall events was associated with soil moisture between 22 and 24%, which corresponded to the conditions of lateral flow generation during the drying periods (Lu et al., 2011; Lv et al., 2013).

During the wetting periods, the angles of the centroid vectors rotated clockwise from a vertical upward direction to lateral upslope and then to a vertical downward direction. The lateral upslope flow occurred in a range of soil moisture between 23 and 24%. The angular sequence for the drying period proceeded from vertical downward to lateral downslope and then vertical upward. The distribution of the angles shown in Fig. 8 and 9 implies that the primary flow path was in the vertical direction, through either downward or upward flow, as was also reported by Kim et al. (2011). The lateral flow, either downslope or upslope, appeared to be a transitional phenomenon during short-term events, such as the wetting periods of rainfall events and the subsequent drying periods. In fact, the change in flow direction during the transitional period under unsaturated conditions has been detected in many studies (Torres et al., 1998; McCord et al., 1991; Jackson, 1992; Pan et al., 1997; Sinai and Dirksen, 2006; Lu et al., 2011;

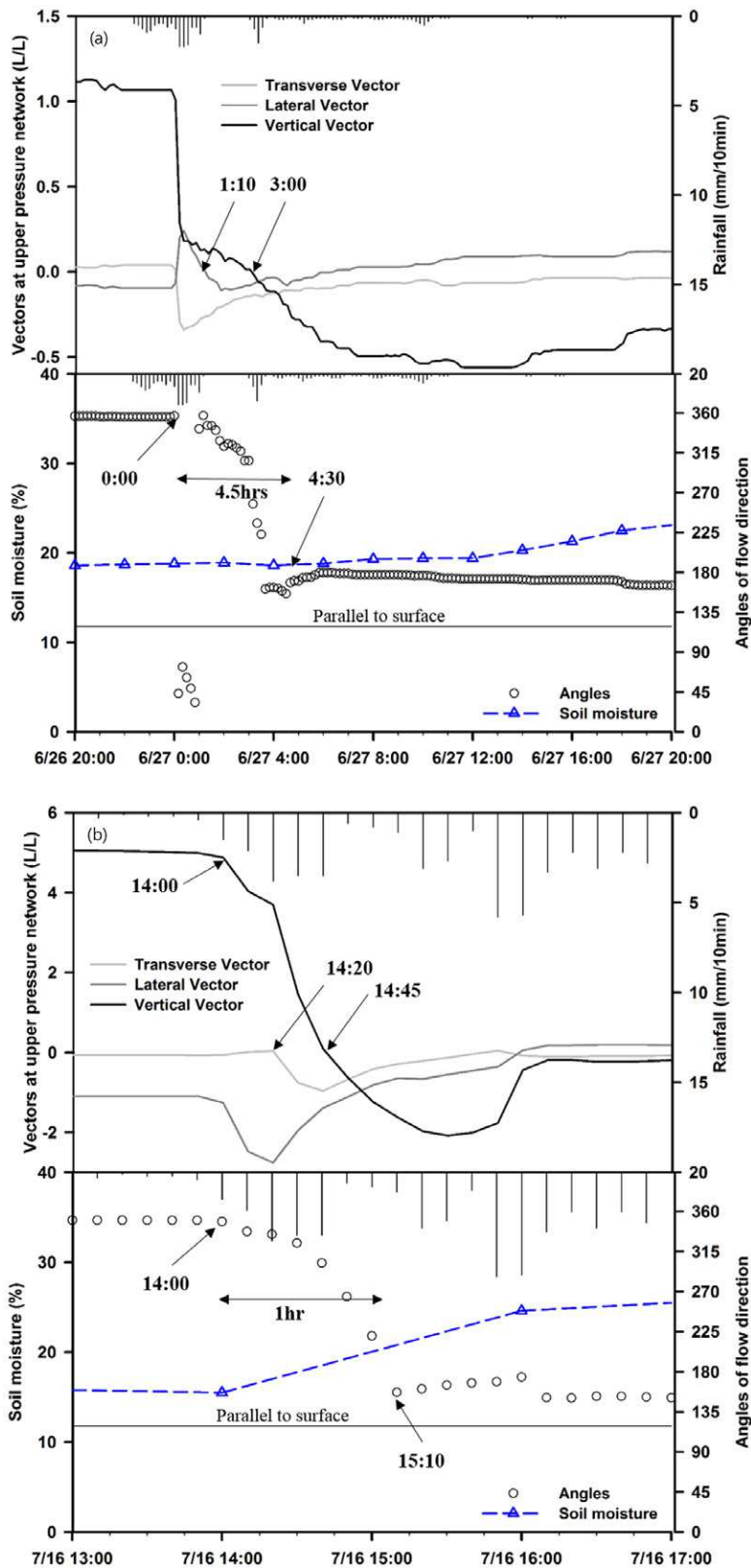


Fig. 8. Transverse, lateral, and vertical components and vector angles of the centroid vector and observed soil moisture in the upper pressure network from (a) 26 to 27 June 2010, and (b) 16 July 2010. The solid lines, open circles, and open triangles indicate the amplitudes of the vectors, the vector angle of the two-dimensional vector (vertical and lateral components), and measured soil moisture, respectively; angles between 360 and 315° are upward flow, those between 315 and 225° are upslope lateral flow, and those between 180 and 135° are downward flow.

Lv et al., 2013). Therefore, the impact of lateral flows (either upslope or downslope) in hillslope hydrology has been shown to be negligible.

Figure 8b shows that the angle of the flow began to change in an upslope lateral direction between 315 and 225° for 20 min (from 1440 to 1500 h on 16 July) as soil moisture increased. Figures 9a and 9b indicate that downslope lateral flow was mainly associated with a drying state. These field monitoring results were consistent with the simulations of a homogeneous and isotropic hillslope (Lu et al., 2011). Figures 8 and 9 confirm that the general parameter determining the flow direction was not the amount of rainfall itself but the state of wetting or drying driven by rainfall on the hillslope (Lu et al., 2011; Lv et al., 2013). Therefore, stationary conditions for non-rainy and rainy periods are associated with upward and downward vertical flows, respectively, while the upslope lateral and downslope lateral flows were generated under transitional conditions for the wetting and drying periods, respectively (Sinai and Dirksen, 2006; Lv et al., 2013). Consequently, the primary hydrological processes are either upward or downward vertical flow, and lateral flow appears to be only temporarily in the switching periods between two major hydrologic components. The proposed method may be useful to identify the temporal distribution of hydrological processes; its extensive applications to more upslope and downslope points may provide direct and meaningful evidence of the spatial and temporal distributions of hydrological processes at a hillslope scale.

Soil Moisture Characteristic Curve and Flow Paths

Application of flow vector analysis can be integrated into the SMCC. The frequency distribution of the flow angle was replaced with multiple pie charts with a soil water tension range of 0 to 200 cm H₂O with an interval of 20 cm H₂O to provide an example of the flow direction transition of this entire range of soil water tension (see Fig. 10). The size of each arrow in Fig. 10 represents the frequency of the flow vector angle in each pie chart.

The results shown in Fig. 10 indicate that vertical upward flow was the primary hydrological process for soil water tensions >80 cm H₂O and that the corresponding soil moistures were <21%. Disturbances in the flow path of vertical upward flow were observed with soil water tensions between 60 and 80 cm H₂O at which the vertical downward flow partially emerged. Major switches in flow paths were found to occur with soil water tensions between 40 and 60 cm H₂O and soil moisture between 22 and 24% (see Fig. 10). These

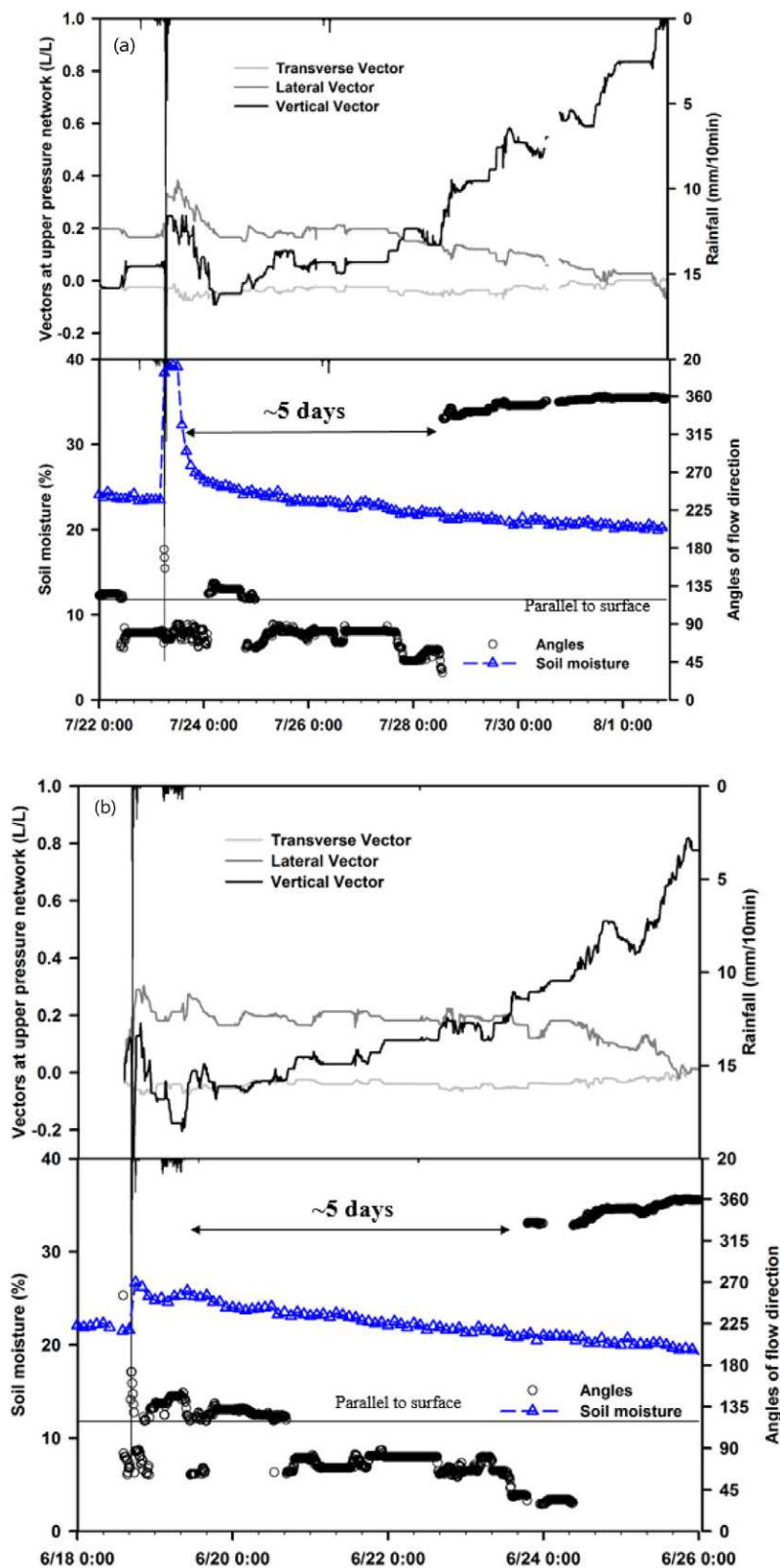


Fig. 9. Transverse, lateral vertical components and vectors angles of the centroid vector and observed soil moisture in upper pressure network from (a) 18 to 26 June 2010, and (b) 22 July 22 to 2 Aug. 2010. The solid lines, open circles, and open triangles indicate amplitudes of vectors, the vector angle of the two-dimensional vector (vertical and lateral components), and measured soil moisture, respectively; angles between 360 and 315° are upward flow, those between 180 and 135° are downward flow, and those between 90 and 45° are downslope lateral flow.

values represents the boundary between the lateral downslope flow ($45^\circ < \theta < 135^\circ$) and the upward exfiltration ($\theta < 45^\circ$, $\theta > 315^\circ$) during the hydrological process. The arrows for wetting and drying processes rotated clockwise, as illustrated in Fig. 8 and 9. Substantial downslope flow components (parallel to the surface) were detected for soil water tensions between 20 and 40 cm H_2O in addition to the horizontal or upward downslope flows that were also shown. A major flow path for soil moisture $>35\%$ and for soil water tension <20 cm H_2O was upward flow in the downslope direction, which may be associated with conditions suitable for return flow (e.g., saturated conditions in the downslope upward direction). The distribution of flow angles in the SMCC highlights that the transitions of hydrological processes may be characterized through measurements of either soil water tension or soil moisture.

Hydrological Implication and Further Applicability

Configurations of hydrological processes have been explored through comparisons of soil moisture, soil water tension, pore pressure, and groundwater level or transit time evaluation of stable water isotopes along hillslope profiles (Torres et al., 1998; Tromp-van Meerveld and McDonnell, 2006; Retter et al., 2006; Sinai and Dirksen, 2006; Kim, 2014; Sprenger et al., 2016). This study provides an alternative and direct method to identify hydrological components in terms of flow vectors at a plot scale. Distributions of flow angles indicate that hydrological processes cannot be completely characterized into existing approaches (e.g., lateral flow and vertical flow) but through four distinct continuous ranges of flow angles. This means that hillslope hydrological processes in the vadose zone can be more accurately described by the probability density function of flow angles (Fig. 7). The proposed method also provides advantages in expressing the temporal variation of hydrological processes. The sequence and duration of flow angle variation can be obtained, which may be useful in understanding the dynamics of hydrological processes at the plot scale. Extensive applications of the proposed method to more upslope and downslope points may provide direct and meaningful evidence of the spatial and temporal distributions of hydrological processes in hillslopes. The combination of SMCC and flow paths may enable feasible identification of flow direction through simple hydrometric measurements at a designated point. Once the relationship between flow paths and SMCC has been established for a specific point, either soil water tension or soil

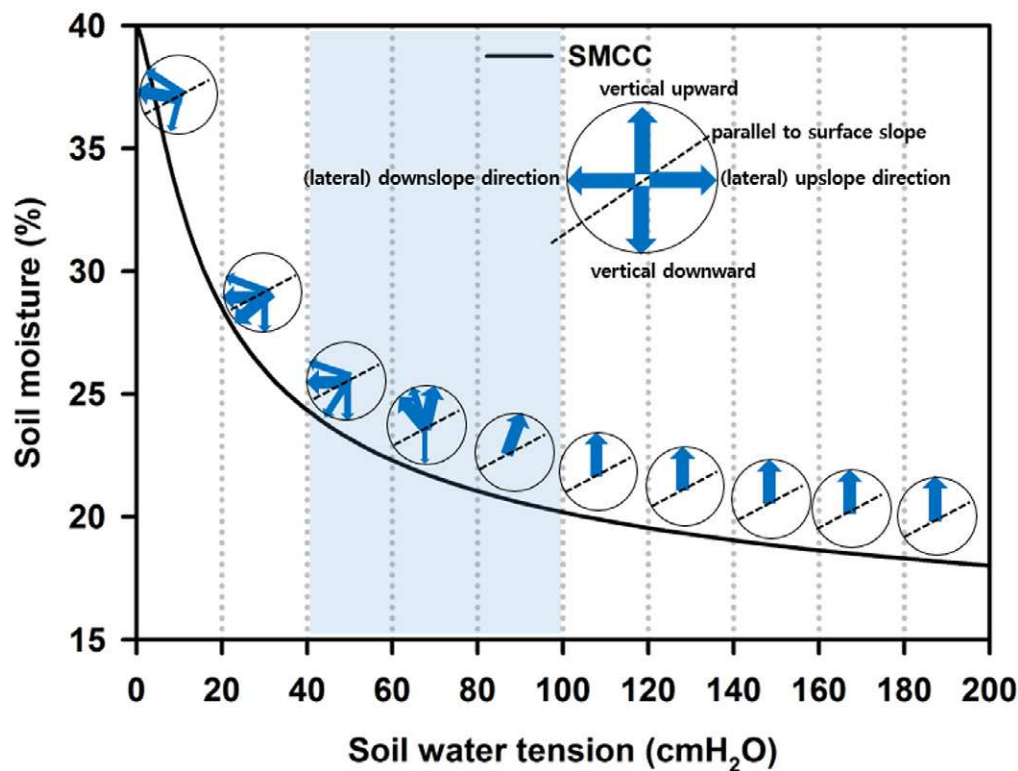


Fig. 10. Schematic of flow paths of the centroid vector denoted on the soil moisture characteristic curve by residual volumetric soil water content $\theta_r = 0.028$, saturated volumetric soil water content $\theta_s = 0.40$, and curve-fitting parameters $\alpha = 0.126 \text{ cm}^{-1}$ and $n = 1.513$, with $R^2 = 0.962$.

moisture measurements provides hydrological implications at the corresponding point. In other words, a hydrometric measurement (either soil moisture or tension) can provide the flow path information. The method developed in this study can be used for a rigorous validation of the three-dimensional modeling approaches of soil water flow (i.e., the dual-porosity model, the dual-permeability model, and the dual-permeability model with immobile water in HYDRUS-2D/3D) [Šimůnek and van Genuchten, 2008]) at a field hillslope. The proposed method can be useful for the field verification of topologically driven hydrological modeling (Quinn et al., 1991; Seibert and McGlynn, 2007). Furthermore, one of the conditions for a hillslope landslide is flow convergence, and the developed method can be useful to predict the time and location for a slope to be triggered to slide and flow downslope (Bogaard and Greco, 2016). During wetting periods, the distribution of angles was bounded between 360° and 135° , indicating a switch from exfiltration to infiltration. If flow angles of disconnected ranges in the drying period (either angles of 135° – 90° or 60° – 360°) were found during wetting periods, a substantial lateral flow was generated, indicating the deterioration of the slope stability. In this study, the transversal flow component was negligible for all periods. If a particular direction from a tensiometer cluster was saturated due to the resistance to downslope drainage (Uchida et al., 2001), a significant transversal flow could be observed, and the original direction of the transversal flow then could be a potential location for slide generation.

Even though flow vector analysis can provide details on a flow path, the cost for data acquisition is substantial (e.g., five points of tensiometer rods are required for the delineation of a flow vector). Reliable operation of pressure networks can be often labor intensive depending on the field condition. Further study using more extensive data sets covering a longer period of time seems to be necessary to obtain a better understanding of hydrological processes. Analysis of flow angles with further consideration of seasonal rainfall patterns (e.g., rainy season and dry season) is also desirable for better characterization of the vadose zone hydrology of a hillslope. The results of this study can be used for other hillslopes if the environmental conditions between the hillslopes are similar (e.g., soil texture, topography, and soil depth). The developed method can be also applied to other study areas as long as the continuity in potential lines within the domain of a pressure network is secured.

Conclusion

We have proposed two methods to delineate various flow gradient vectors using an in situ pressure sensor network. The total vector analysis denoted pressure gradients for the volume of a quadrangular pyramid, whereas the centroid vector indicated pointwise pressure components to minimize the biases from a cluster of pressure sensors with asymmetrical geometry. Both methods provided time series of three-dimensional pressure flow components and represented fine-scale transitional hydrological processes between vertical flow and lateral flow.

Both time series of the three-dimensional flow vectors and the resultant two-dimensional vectors indicated that the angles of flow changed from vertical upward to lateral upslope followed by vertical downward flow during a wetting period. However, the flow direction changed during the drying period from vertical downward to lateral downslope and then to vertical upward. Probability density distributions of the flow vectors were used to quantify the hydrological components such as downward flow, upward flow, downslope lateral flow, and upslope lateral flow. The proposed method was useful for characterizing hydrological processes at a point scale. The integration of flow vector analysis into the SMCC was used to identify the transition of the hydrological processes that ranged from upward flow to downward flow, downslope lateral flow, and return flow depending on either soil water tension or soil moisture measurements. The abundance of macropores and intense root developments at shallow depths was associated with diverse flow directions at the upper pressure network. Weathered rocks and gravel at the deeper soil depth were responsible for dominance in the vertical flow at the lower pressure network. The spatial and temporal distributions of flow vectors along the hillslope, and the response patterns of their relationships with soil moisture, will be useful for addressing research problems such as the field verification of soil water flow modeling with preferential flow, the dual-permeability module in HYDRUS-3D, and slope stability analysis during rainfall events.

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